Lucien S. Carroll

Voice/NLP Engineer · Data Development and Evaluation

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PROFILE

- Technical leader with 10+ years shipping and evaluating production NLP systems.
- Designed and ran evaluation for LLM and classifier features: automated evaluation, human evaluation, failure analysis, and quality control across 12+ languages.
- Built data development pipelines from research prototype through production launch: acquisition, data synthesis, annotation, and quality control.
- Shipped the voice interface for Cisco meeting room devices; designed in-situ wakeword data collection, ran ASR + NLU evaluations, and delivered iterative improvements.
- Coordinated across product, engineering, domain experts, and business operations — turning under-specified requests into shipped high-quality features.

EXPERIENCE

Cofounder, Head of Data/ML

2025 Dec–2026 Aug

Kedara.care, California

- Built the full stack (React frontend, Python backend, and Docker + GitHub Actions deployments) for the Kedara Guide assistant, enabling health and wellness resource discovery for eldercare.
- Designed evaluation frameworks and instrumented Kedara Guide with eval hooks and OpenTelemetry traces for observability and improvements of the RAG + dialogue pipeline.
- Established automation and guidelines for software development best practices.

NLP Technical Leader

2023 Sep–2025 Nov

Cisco Systems, San Francisco, CA

- Researcher, software engineer, and advisor for prototyping, data development, evaluation, and responsible AI review of language features based on OpenAI and open-weight LLMs.
- Led implementation of Webex Meeting summaries features (automated chapters, notes, action items) from development through to adoption by millions of Webex users globally.
- Extended Meeting summaries to 12+ languages, refactoring production services and automated evaluation processes for multilingual support.
- Designed and advised data development and human-in-the-loop, PII-respecting evaluation processes for other features, e.g. Q&A, cross-meeting search, and Vidcast summaries.

Data Science Team Leader

2019 Apr–2023 Sep

Cisco Systems, San Francisco, CA

- Led data development, evaluation, and product metrics efforts for the Webex Assistant for Rooms, the voice interface for Cisco Room devices.
- Instigated and led internationalization of Webex Assistant, expanding classifier-based NLU systems and template-based NLG systems from English to 7 languages.
- Managed and mentored an international team of data scientists and linguists, overseeing data acquisition, data synthesis, annotation, and quality control for training and evaluating multilingual NLP pipelines.
- Set up vendor contracts for data suppliers and contract workers, projected costs, generated budget requests, and approved invoices.

NLP Engineer

2017 May–2019 Apr

Cisco Systems, San Francisco, CA

- Designed and implemented custom data collection process and tool for in-situ wakeword recordings, used to seed training and evaluation of keyword detection edge models.
- Designed and ran evaluation of ASR providers for Webex Assistant.

Search Quality Engineer

2015 Mar–2017 May

MindMeld, San Francisco, CA

- Served as tech lead for production contract, guiding conversational assistant from POC through launch: articulated solutions, oriented new team members, and performed key development and chores to maintain and improve system behavior.
- Innovated core features and error-analysis and diagnostic tools for intent classification and named entity recognition in the query parsing pipeline.

Graduate Research Assistant

2010 Jun–2014 Jul

UCSD Linguistics, San Diego, CA

- Developed lexicon and corpus systems for the Rarámuri Documentation Project, integrating FLEx lexical data and Praat acoustic analysis with ELAN text annotations.
- Collaborated on fieldwork, research design, data analysis, publications, and presentations of research at professional conferences and workshops.
- Trained and supervised undergraduate research assistants.

EDUCATION

2007–2015	Ph.D. Linguistics, UCSD, San Diego, CA
2003–2006	M.A. program, Computational Linguistics, SDSU, San Diego, CA
1997–2002	B.S. Physics and B.A. Linguistics, UCSB, Santa Barbara, CA

PROJECTS & COMMUNITY WORK

2021–2022	Fine-tuned XLSR/XLS-R speech models for Kinyarwanda, Luganda, Uzbek, Uyghur, and Kyrgyz in Hugging Face community events.
2019–2026	Reviewer for ACL conferences (ARR, EMNLP, COLING, LREC, and others).
2023–2025	Reviewer for INTERSPEECH.

SKILLS

Expertise	AI product engineering, LLM/generative AI systems, data development and annotation, evaluation frameworks, team leadership and mentoring, machine learning, question answering, named entities, search ranking, dialogue systems, statistics, product metrics.
Tools/platforms	Python (Scikit-Learn, NLTK, PyTorch, spaCy, Streamlit), R/RStudio, Praat, UNIX shell, Prolog, JavaScript, Git, AWS, GCP, Elasticsearch, SQL, LaTeX, Hugging Face.
Languages	English, Spanish (professional), Mandarin and Portuguese (moderate).